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Scientific Machine Learning and Numerical Methods in Derivative Pricing

Zihan Liang, Qimu Luo, Xiaoya Wu, Ruixi Xu

Supervisor: Dr. Ren Zhao, Dr. Yiru Ye

Department of Foundational Mathematics, School of Mathematics and Physics

西交利物浦大學 Xi'an Jiaotong-Liverpool University

Abstract

This project develops and evaluates numerical methods and physics-informed neural networks (PINNs) for free-boundary American-option risk surfaces under the Black–Scholes framework, with hybrid neural extensions to high-dimensional states. Early exercise leads to a variational-inequality (VI) problem in which option value, exercise boundary, and price sensitivities must be resolved consistently. We compare Crank–Nicolson (CN) linear complementarity problem (LCP) solvers, construct a high-accuracy diagonally implicit Runge–Kutta (DIRK)–Policy Iteration–sinh reference, and study low-dimensional obstacle PINNs and high-dimensional hybrid-VI models. Brennan–Schwartz projected LU achieves a 26.0× speedup over projected successive over-relaxation (PSOR), while the Structure-aware PINN achieves a 26.7× reduction in price root-mean-square error (RMSE) relative to the Vanilla PINN. Overall, structural enforcement may improve neural accuracy, although reliable free-boundary recovery remains more challenging than price approximation.

Model Problem

European Pricing and Black–Scholes PDE

For asset price S, time-to-maturity τ, risk-free rate r, dividend yield q, and volatility σ, define

$$\mathcal{L}_{\mathrm{BS}} V = \tfrac{1}{2}\sigma^2 S^2 V_{SS} + (r-q) S V_S - rV.$$

European options satisfy:

$$V_\tau - \mathcal{L}_{\mathrm{BS}} V = 0, \qquad V(S,0) = \Phi(S).$$

The European call has a closed-form Black–Scholes solution, providing an analytical reference for validating the numerical methods.

American Free-Boundary Formulation

For an American payoff $\Phi(S)$, early exercise introduces the obstacle $V(S,\tau) \geq \Phi(S)$. The option value satisfies the variational inequality

$$\min(V - \Phi,\ V_\tau - \mathcal{L}_{\mathrm{BS}} V) = 0.$$

$$\underbrace{V = \Phi}_{\text{exercise}} \ \Big|\ \underbrace{S = S^*(\tau)}_{\text{free boundary}} \ \Big|\ \underbrace{V_\tau - \mathcal{L}_{\mathrm{BS}} V = 0}_{\text{continuation}}$$

American Exercise Regimes

American put $\Phi_{\mathrm{put}} = (K-S)^+$ low-S exercise region.
Dividend-paying call $\Phi_{\mathrm{call}} = (S-K)^+$ high-S exercise.
No-dividend call $q = 0 \Rightarrow C_A = C_E$ no early exercise.

Key Outputs

$$V(S,\tau), \quad S^*(\tau), \quad \Delta = V_S, \quad \Gamma = V_{SS}$$

Outputs capture option value, exercise behavior, and local price sensitivity.

Methodology

Numerical PDE Methods

N1. Strict CN–LCP Solvers

LCP solver efficiency

Crank–Nicolson discretization converts the continuous American variational inequality into discrete linear complementarity problems, one at each time step[2]:

$$V^{n+1} \geq g, \qquad A_{\mathrm{CN}} V^{n+1} - b^n \geq 0,$$
$$(V^{n+1} - g)^{\mathsf{T}} (A_{\mathrm{CN}} V^{n+1} - b^n) = 0.$$

CN–Projected SOR[2]

Basic Classical Benchmark

$$V_i^{k+1} = \max\Big(g_i,\ V_i^{k} + \omega(\tilde{V}_i^{k+1} - V_i^{k})\Big).$$

Projected SOR applies a relaxed Gauss–Seidel candidate and projects it back to the payoff obstacle at every node.

CN–Brennan–Schwartz (Projected LU)

Accelerated Strict Solver

Brennan–Schwartz exploits the tridiagonal CN system through LU factorization and projected backward substitution[1].

N2. High-Accuracy Reference

Discretization accuracy for boundary and Greeks

The selected reference combines **L-stable second-order DIRK** time stepping, stage-wise LCP **Policy Iteration**, and a **strike-concentrated $sinh$-mapped** spatial grid.

Algorithm 1: DIRK–Policy–$sinh$

Input: Contract/PDE data; M, N, ε
Output: V, $S^*(\tau)$, Δ, Γ
Spatial: Strike-concentrated $sinh$ grid
Concentrate nodes near K and assemble $\mathcal{L}_h$
Temporal: L-stable DIRK[3]
Set $\theta = 1 - 1/\sqrt{2}$; initialize $V^0 \leftarrow \Phi$ with two BE damping starts
for *DIRK time step* n **do**
  for *implicit stage* $i = 1, 2$ **do**
    Assemble $(I - \theta \Delta\tau_n \mathcal{L}_h) Y_i = R_i$
    Stage solver: Policy Iteration
    repeat
      Update policy and solve the induced linear system
    until *stage residual* $\leq \varepsilon$
  $V^n \leftarrow Y_2$
Extract $S^*(\tau)$, Δ, Γ and return V

Physics-Informed & Hybrid Neural Methods[4]

The neural models approximate the option value by V_Θ. Define $G_\Theta = V_\Theta - \Phi$ and $\mathcal{R}_\Theta = V_{\Theta,\tau} - \mathcal{L} V_\Theta$, where $\mathcal{L}$ is the corresponding one- or d-dimensional Black–Scholes operator. Thus complementarity is: $G_\Theta \geq 0, \quad \mathcal{R}_\Theta \geq 0, \quad G_\Theta \mathcal{R}_\Theta = 0.$

P1. Low-Dimensional Obstacle PINNs

We use log-moneyness $x = \log(S/K)$ and normalized time $s = \tau/T$.

Vanilla soft-LCP PINN.

The network learns V_Θ directly; terminal, boundary, obstacle and complementarity conditions enter as soft penalties.

Structure-aware ETC+FB PINN.

Using $V_\Theta = V_{\mathrm{lift}} + s N_\Theta$ enforces the terminal $V_\Theta(S,0) = \Phi(S)$ exactly. Fischer–Burmeister (FB)[5] uses $\phi_{\mathrm{FB}}(a,b) = \sqrt{a^2+b^2} - a - b$, where $\phi_{\mathrm{FB}}(G_\Theta, \mathcal{R}_\Theta) = 0$ encodes complementarity. A smooth approximation is used in training.

P2. High-Dimensional Hybrid VI

Define the d-dimensional state $\mathbf{x} = (x_1, \ldots, x_d)$, with $d \in \{10, 20, 50\}$. $V_\Theta = \Phi + s\ \mathrm{softplus}(N_\Theta)$ guarantees the obstacle $V_\Theta \geq \Phi$.

For training, both models learn from reference prices, while the FB residual enforces the PDE and exercise constraints.

FC-HVI: a fully connected (FC) network takes the state $\mathbf{x}$ and time s as inputs.

DeepSets-HVI network[6]: mean pooling forms $z = \frac{1}{d}\sum_{i=1}^{d} \varphi_\Theta(x_i)$, making z invariant to asset ordering before the final value prediction.

Results

R1. LCP Solver Efficiency

Under the same CN–LCP discretization and 10^{-12} residual tolerance, Projected LU reduces the median runtime from 2.44×10^{-1} s to 9.38×10^{-3} s, corresponding to a **26.0× speedup** over CN–PSOR.

R2. High-Accuracy Reference: Accuracy & Convergence

Price Accuracy vs 50k-Step Binomial-Tree

Method	Price RMSE	Max |Price Error|
CN + PSOR	5.01×10^{-5}	2.71×10^{-4}
DIRK + Policy + $sinh$	$\mathbf{2.70 \times 10^{-6}}$	$\mathbf{1.01 \times 10^{-5}}$

American put: $K = 1,\ T = 1,\ r = 0.05,\ q = 0.10,\ \sigma = 0.60.$

DIRK–Policy–$sinh$ achieves higher **price accuracy**, with an **18.5× reduction** in RMSE.

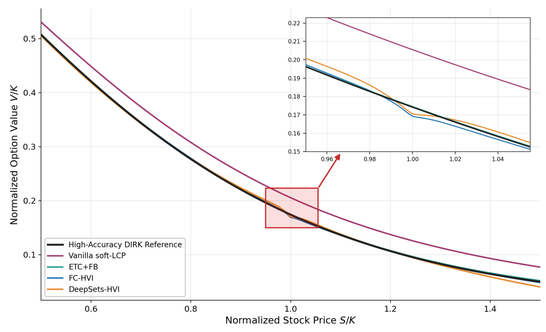
Temporal and Spatial Refinement Convergence

Temporal refinement – DIRK			**Spatial refinement – sinh-grid**		
(M, N)	Price RMSE	Γ RMSE	(M, N)	Price RMSE	Γ RMSE
$(240, 120)$	6.35×10^{-8}	2.75×10^{-5}	$(120, 960)$	1.10×10^{-5}	1.92×10^{-3}
$(240, 240)$	1.47×10^{-8}	5.18×10^{-6}	$(240, 960)$	2.64×10^{-6}	3.93×10^{-4}
$(240, 480)$	3.00×10^{-9}	1.37×10^{-6}	$(480, 960)$	4.70×10^{-7}	1.03×10^{-4}

American put: $K = 1,\ T = 0.25,\ r = 0.01,\ q = 0,\ \sigma = 0.20.$

The observed orders are approximately 1.9 – 2.5, consistent with second-order behavior and showing **regular, convergence**. These two tables support DIRK–Policy Iteration–$sinh$ as the high-accuracy reference for the PINNs comparison below.

R3. PINN Evaluation



American-put price at $\tau/T = 0.5$

PINNs Price RMSE against the high-accuracy reference

Model	Price RMSE
Vanilla PINN	2.79×10^{-2}
Structure-aware PINN	$\mathbf{1.05 \times 10^{-3}}$
FC-HVI ($d = 50$)	2.31×10^{-3}
DeepSets-HVI ($d = 50$)	4.00×10^{-3}

Common domain:
$S/K \in [0.5, 1.5], \tau/T \in [0.01, 1].$

Structure-aware PINN achieves a **26.7× reduction** in price RMSE relative to the Vanilla PINN and the lowest error among the neural models. The $d = 50$ HVI models remain at the 10^{-3} RMSE scale, but do not surpass the Structure-aware PINN in this comparison.

R4. Free Boundary Recovery

Against the binomial-tree benchmark, DIRK reduces boundary RMSE from CN–PSOR 9.40×10^{-3} to $\mathbf{2.39 \times 10^{-3}}$. Vanilla soft-LCP detects only **0.6%** of boundary crossings, versus **28.2%** for ETC+FB, reflecting the difficulty of free-boundary recovery for PINNs.

Conclusion

Method design matters for American-option risk surfaces. For a fixed CN discretization, Projected LU substantially improves LCP-solver efficiency, while DIRK–Policy–sinh provides a high-accuracy reference with regular refinement behavior. Neural performance depends strongly on how the free-boundary structure is represented: ETC+FB substantially improves the vanilla PINN, while hybrid-VI models extend the framework to high-dimensional states. However, accurate price approximation does not guarantee reliable free-boundary recovery. Future work will focus on improving neural boundary and Greek recovery and extending matched validation across broader high-dimensional and dividend-paying call settings.

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